ML-Based Cooling Technique Recommendation — Methodology

Dataset Generation. Scenarios are produced via Monte Carlo sampling across six features: outdoor temperature, relative humidity, IT load, electricity tariff, water tariff, and grid carbon intensity. Each scenario is run through all three cooling simulators (Air Economizer, Evaporative, Chilled Water) with identical inputs. The label `bestTechnique` is assigned to the feasible technique with minimum energy consumption. Per-technique outcome columns are excluded from model inputs to prevent leakage.

Model Training. A scikit-learn Pipeline applies median imputation and StandardScaler before a `RandomForestClassifier` with `class_weight='balanced'`. The dataset is split 80/20 with stratification. RandomizedSearchCV over 20 iterations optimises Macro-F1 via 5-fold cross-validation. The final model is evaluated on accuracy, Macro-F1, confusion matrix, and feature importances, with robustness confirmed over 30 repeated stratified splits. The artifact is saved to `cooling_recommender_rf.pkl`.

Inference Pipeline. When a user completes any one simulation, the system reads real annual cost, emissions, water, energy, PUE, and 8,760 hourly rows from the database. Annual averages of temperature, humidity, and IT load are computed and combined with tariff settings to form the six-feature ML input. Since only one technique was simulated, the other two are estimated using climate-aware dynamic efficiency factors — penalty variables derived from real site temperature and humidity scale each technique's cost, emissions, water, and energy relative to the current simulation's real values. Feasibility is checked against physical thresholds (Air Economizer fails above 27°C or 80% RH; Evaporative fails above 70% RH). The Random Forest predicts the best technique; if infeasible, the lowest composite-score feasible alternative is selected (score = 0.5×cost + 0.3×emissions + 0.2×water, normalised). Real metrics, failure findings, and comparisons are sent to Groq LLM (`llama-3.3-70b-versatile`) which generates an 8–10 sentence justification and a 5-year future impact paragraph. The response — recommendation, justification, future impact, and comparison table — is stored in the database and rendered in the frontend Recommendations tab.

Limitations. Efficiency factors are engineering approximations, not empirically calibrated. Estimated metrics for non-simulated techniques are approximations; exact values require running all three simulations with identical inputs.